observable universe, mostly photons and neutrinos. That is a number much greater than two. And each particle is located in three-dimensional space, not just a one-dimensional line. How in the world are we supposed to visualize a wave function that assigns an amplitude to every possible configuration of 10^{88} particles distributed through three-dimensional space?

We're not. Sorry. The human imagination wasn't designed to visualize the enormously big mathematical spaces that are routinely used in quantum mechanics. For just one or two particles, we can muddle through; more than that, and we have to describe things in words and equations. Fortunately, the Schrödinger equation is straightforward and definite in what it says about how the wave function behaves. Once we understand what's going on for two particles, the generalization to 10^{88} particles is just maths.

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The fact that wave functions are so big can make thinking about them a little unwieldy. Happily we can cast almost everything interesting to say about entanglement into the much simpler context of just a few qubits.

Borrowing from a whimsical tradition in the literature on cryptography, quantum physicists like to consider two people named Alice and Bob who share qubits with each other. So let's imagine two electrons, *A* belonging to Alice and *B* belonging to Bob. The spins of those two electrons constitute a two-qubit system, and are described by a corresponding wave function. The wave function assigns an amplitude to each configuration of the system as a whole, with respect to something we might observe about it, such as its spin in the vertical direction. So there are four possible measurement outcomes: both spins are up, both spins are down, *A* is up and *B* is down, and *A* is down and *B* is up. The state of the system is some superposition of these four possibilities, which are the basis states. Within each set of parentheses, the first spin is Alice's, and the second is Bob's.